

EXP 02 Candidate Elimination Algorithm

```
import pandas as pd
import numpy as np

# Student Placement Dataset
data = {
    'CGPA': ['High', 'High', 'Medium', 'High', 'Medium', 'High'],
    'Communication': ['Good', 'Average', 'Good', 'Good', 'Average', 'Excellent'],
    'Internship': ['Yes', 'Yes', 'No', 'No', 'Yes', 'Yes'],
    'Placement': ['Yes', 'Yes', 'No', 'Yes', 'No', 'Yes']
}

df = pd.DataFrame(data)

print("Register Number : 192321113")
print("\nTraining Data:")
print(df)

concepts = np.array(df.iloc[:, :-1])
target = np.array(df.iloc[:, -1])

# Initialize Specific Hypothesis
S = concepts[0].copy()

# Initialize General Hypothesis
G = [["?" for i in range(len(S))]]

# Candidate Elimination Algorithm
for i in range(len(target)):
    if target[i] == "Yes":
        for j in range(len(S)):
            if S[j] != concepts[i][j]:
                S[j] = "?"
    else:
        for j in range(len(S)):
            if S[j] != concepts[i][j]:
                G.append(["?" if x != j else S[j] for x in range(len(S))])

print("\nSpecific Hypothesis (S):")
print(S)
```

```
print("\nGeneral Hypothesis (G):")
for g in G:
    print(g)
```

```
*** Register Number : 192321113

Training Data:
  CGPA Communication Internship Placement
0   High           Good           Yes      Yes
1   High           Average        Yes      Yes
2   Medium          Good           No       No
3   High           Good           No       Yes
4   Medium          Average        Yes      No
5   High           Excellent       Yes      Yes

Specific Hypothesis (S):
['High' '?' '?']

General Hypothesis (G):
['?', '?', '?']
['High', '?', '?']
['?', '?', '?']
['?', '?', 'Yes']
['High', '?', '?']
['?', '?', '?']
['?', '?', '?']
```